LISA ORII

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SUMMARY

I apply methods from human-computer interaction to guide human-centered digital health in low- and middle-income countries to improve patient and provider experiences. I partner with governments and implementing partners, and work with health experts, designers, and engineers to design, develop, and evaluate digital technologies used in real-world clinical contexts.

EDUCATION

University of Washington Sep 2021 – Jun 2026 (Expected) | Seattle, WA, USA
Ph.D. in Computer Science & Engineering
Graduate Certificate in Global Health of Women, Adolescents, and Children
- Proposed Thesis: Human-Centered Mobile Health in Resource-Constrained Settings

Wellesley College Sep 2017 – Jun 2021 | Wellesley, MA, USA
Bachelor of Arts in Computer Science and Philosophy
- Honors: Phi Beta Kappa, Sigma Xi, *magna cum laude*

EXPERIENCE

University of Washington – Graduate Research Assistant Sep 2021 – Present | Seattle, WA, USA
Contraceptive Decision-Support App for Adolescent Girls and Young Women – Kenya
- Led a team of 13 engineers, designers, and health experts to manage a decision-support app from concept to launch.
- Drove a 3-month launch across 6 sites in Kenya with 100 users, yielding 94% confidence in contraceptive choice.
- Ran participatory design sessions and usability testing with 45 stakeholders to inform user-centered product strategy.
- Developed a responsible AI design roadmap with 9 use cases by conducting chatbot concept testing 45 stakeholders
- Tested chatbot concepts with 45 stakeholders, defining 6 priority use cases and responsible AI designs.

Security of Electronic Medical Record Systems for HIV Care – Malawi
- Partnered with a public HIV care program in Malawi to identify security gaps in electronic medical record systems.
- Led field research and engagement with 75 stakeholders to understand security concerns and system constraints.
- Synthesized qualitative data in ATLAS.ti into guidance for patient data protection, informing system design and use.
- Presented insights and recommendations to 9 program decision-makers, facilitating policy and strategic discussions

Clinical Decision-Support App for Children's Respiratory Illnesses – Uganda
- Developed focus group protocols to capture provider experiences and adoption barriers during a 3-month pilot.
- Conducted qualitative analysis in ATLAS.ti to identify barriers and facilitators to adoption and workflow integration.

Reliability of Digital Financial Services – Nigeria
- Designed a national longitudinal study on digital financial service in Nigeria to inform consumer interest protection.
- Analyzed 49,000 transactions in Python and R, using statistical analysis to identify transaction failure patterns.
- Produced technical briefs with data visualizations to inform national stakeholders on service reliability gaps.

OMRON SINIC X Corporation – Research Intern Jun 2025 – Sep 2025 | Tokyo, Japan
- Designed and ran a 60-person mixed-methods study on AI-mediated communication for peer support.
- Iteratively built an LLM-powered system that generated context-aware, empathetic reactions to user messages.
- Conducted analysis using ATLAS.ti and regression modeling in R to assess AI impacts on communication.
- Developed a framework for context-aware, risk-sensitive AI that mediates paralinguistic communication.

PATH – Intern Apr 2024 – Sep 2024 | Seattle, WA, USA
- Defined 250 data elements and 16 personas for integrated newborn nutrition following WHO SMART Guidelines.
- Identified 10 improvement areas in advocacy, data traceability, and quality control by analyzing 100+ expert inputs.
- Contributed human-centered research methods to a concept note shared with funders, informing program design.

The University of Tokyo – Research Intern Apr 2020 – May 2022 | Remote

- Led 18-participant mixed-methods study to evaluate interactions with a speech analysis software.
- Conducted qualitative analysis and statical analysis in R to assess user experiences with speech manipulation.
- Defined 3 design features of user-controlled speech training systems that enhance user confidence.

Wellesley College – Undergraduate Research Assistant Aug 2020 – Sep 2021 | Wellesley, MA, USA

- Web scraped 2,900+ social media posts to explore attitudes and experiences with AI in the trucking industry.
- Conducted qualitative analysis and time-series modeling to identify emerging trends.

MIT Media Lab – Undergraduate Research Assistant Feb 2019 – Nov 2020 | Cambridge, MA, USA

Water Ecosystem Monitoring Kit – Benin

- Engineered a water-quality sensor kit, extending battery life from 3 hours to 2 weeks with solar batteries.
- Integrated wireless data transmission, enabling real-time monitoring and greater usability of field operations.

Well-Being Urban Indicators at the District Level

- Conducted literature review of 140 articles and country resources to explore definitions of well-being.
- Defined 5 urban indicators and 13 urban attributes that measure the impacts of urban ecosystems on well-being.

PEER-REVIEWED PUBLICATIONS

JOURNAL ARTICLES

J.4 **"They gave us the right to choose." A qualitative study of preferences for differentiated service delivery location among recipients of antiretroviral therapy at Lighthouse Trust in Lilongwe Malawi**
Lisa Orii, Kate S Wilson, Jacqueline Huwa, Christine Kiruthu-Kamamia, Odala Sande, Agness Thawani, Astrid Berner-Rodoreda, Evelyn Viola, Hannock Tweya, PetrosTembo, Wapu Masambuka, Richard Anderson
PLOS ONE

J.3 **eHealth Data Security and Privacy: Perspectives from Diverse Stakeholders in Malawi**
Lisa Orii, Caryl Feldacker, Hannock Tweya, and Richard Anderson
Proceedings of the ACM on Human-Computer Interaction

J.2 **Menopause Legacies: Designing to Record and Share Experiences of Menopause Across Generations**
Shaan Chopra, **Lisa Orii**, Katherine Juarez, Nussara Tieanklin, James Fogarty, and Sean A. Munson
Proceedings of the ACM on Human-Computer Interaction

J.1 **Methodology for Establishing Well-Being Urban Indicators at the District Level to be Used on the CityScope Platform**
Lisa Orii, Luis Alonso, Kent Larson
Sustainability

CONFERENCE PAPERS & ABSTRACTS

C.8 **Implementing a novel mobile health intervention to provide pharmacy-based, person-centered contraceptive decision support for youth in Kenya: A mixed-methods feasibility**
Elizabeth K. Harrington, Dismas Congo Ouma, **Lisa Orii**, Merceline Awuor, Cellestine Aoko, Richard Anderson, Serah Gitome, Elizabeth A. Bukusi
2025 International Conference on Family Planning (ICFP)

C.7 **Supporting Contraceptive Decision-Making in the Intermediated Pharmacy Setting in Kenya**
Lisa Orii, Elizabeth K Harrington, Serah Gitome, Nelson Kiprotich Cheruiyot, Elizabeth Anne Bukusi, Sandy Cheng, Ariel Fu, Khushi Khandelwal, Shrimayee Narasimhan, and Richard Anderson
Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems

C.6 **HIV Client Perspectives on Digital Health in Malawi**
Lisa Orii, Caryl Feldacker, Jacqueline Madalitso Huwa, Agness Thawani, Evelyn Viola, Christine Kiruthu-Kamamia, Odala Sande, Hannock Tweya, and Richard Anderson
Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems

C.5 **"Designing for Speech Practice Systems: How Do User-Controlled Voice Manipulation and Model Speakers Impact Self-Perceptions of Voice?**
Lisa Orii, Nami Ogawa, Yuji Hatada, and Takuji Narumi
Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems

C.4 **Perceptions of Trucking Automation: Insights from the r/Truckers Community**
Lisa Orii, Diana Tosca, Andrew L Kun, and Orit Shaer
13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications

C.3 **Perceptions on the Future of Automation in r/Truckers**
Lisa Orii, Diana Tosca, Andrew L Kun, and Orit Shaer
Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems

C.2 **Hooks in the Headline: Learning to Generate Headlines with Controlled Styles**
Di Jin, Zhijing Jin, Joey Tianyi Zhou, **Lisa Orii**, and Peter Szolovits
Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics

C.1 **Co-Designing an Environmental Observatory to Support Eco-Entrepreneurship in Benin**
Danielle Wood, Fohla Mouftaou, Temilola Fatoyinbo, David Lagomasino, Ufuoma Ovienmhada, Eric Ashcroft, Lelia Hampton, **Lisa Orii**, TojumiOluwa Adegboyega, Anisha Nakagawa and Julliet Wanyiri
AGU Fall Meeting Abstracts

HONORS, AWARDS, FELLOWSHIPS

Japan Student Services Organization (JASSO) Graduate Scholarship	2025
ACM CHI 2024 Best Paper Honorable Mention (top 5%)	2024
Wellesley College Horton-Hallowell Fellowship	2025
Quad Fellowship	2022
MIT Technology Review's Innovators Under 35 Japan	2022
Funai Overseas Scholarship	2020
Grace Hopper Celebration 2020 Scholarship	2020
Japan Student Services Organization (JASSO) Undergraduate Scholarship	2017
Grew Bancroft Foundation Scholarship	2017
Kanazawa Iwao Memorial Foundation Scholarship	2017

GRANTS

University of Washington Global Innovation Fund Tier 1 Research Award ($5,000)	2025
University of Washington Global Innovation Fund Tier 1 Research Award ($20,000)	2023

TEACHING

University of Washington

Data Programming, Introductory Python (90 Undergraduates)	Winter 2025, Autumn 2025, Winter 2026
Computing for Social Good (30 Graduates)	Autumn 2024
Capstone Software Design for Global Health (25 Undergraduates)	Spring 2024
Human-Computer Interaction (40 Undergraduates)	Winter 2024

Wellesley College

Human-Computer Interaction (30 Undergraduates)	Spring 2021
Human-Automation Interaction (30 Undergraduates)	Spring 2021

ACADEMIC SERVICE

Reviewing

Proceedings of the ACM on Human-Computer Interaction	2023 – Present
ACM CHI Conference on Human Factors in Computing Systems	2023 – Present

University Organization

University of Washington Change Seminar Co-Organizer	2023 – 2024

University of Washington ICTD Seminar Co-Organizer 2023 – 2024
University of Washington CSE Prospective Student Committee Co-Chair 2023 – 2024
University of Washington Pre-Application Mentorship Service Volunteer 2021 – 2024

Conference Organization
ACM CHI Conference Student Volunteer 2025 – 2026
ACM COMPASS Student Volunteer 2022
ICTD Student Volunteer Chair 2022
Automotive UI Student Volunteer 2021

MENTORSHIP

Krisha Khandelwal
Ariel Fu
Khushi Khandelwal
Shrimayee Narasimhan
Shobhit Srivastava